CALCULUS · UNIT 1: PRECALCULUS REVIEW

Phone Battery Lab

Review function families by modeling a phone's battery, charging, and app data.

Honors

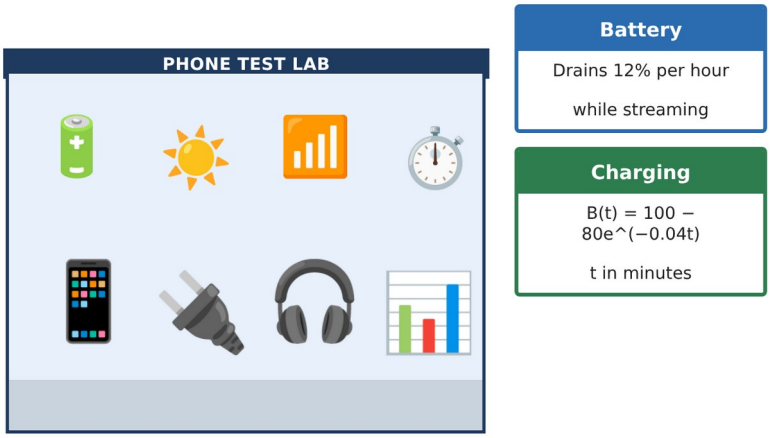
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A phone company wants a student lab to test its new phone. The battery drains while streaming, charges quickly at first and then slowly, the screen brightness follows the sunlight, and the data plan has a price that changes after a limit.

You will use the function families you need for calculus to model the phone and answer the company's questions.



Learning Goals

- I can model with function families and analyze their behavior.
- I can use rates of change and limits informally.
- I can justify model choices with data and reasoning.

Project Timeline

When	What you will do
Day 1	Model Analysis
Day 2	Composite Models
Day 3	Brightness & Rates
Final	Battery Lab Report + reflection

Part 1: Model Analysis

Day 1 · Use the Lab Data Cards.

Lab Data Cards

Streaming	Gaming	Charging
Starts at 95%	Loses 18% of the	B(t) = 100 − 80e^(−0.04t)
loses 12% per hour	remaining charge per hour	

Data Plan	Brightness	Temperature
$30 up to 15 GB	Max 900 nits at 1 p.m.	F = 1.8C + 32
+$4 per GB over	Min 100 nits at 1 a.m.	

1. Show algebraically that B(t) never reaches 100.

2. Find the time for B to go from 50% to 90% and compare with 20% to 50%.

3. Estimate the instantaneous rate of change of B at t = 30 using a symmetric difference with h = 0.1.

4. As h → 0, what value do the difference quotients for B at t = 0 approach? Use h = 1, 0.1, 0.01.

Part 2: Composite Models

Day 2 · Combine functions.

5. The phone's temperature in °C while charging is $C(B) = 25 + 0.12B$. Find $C(B(t))$ and its value at $t = 60$.

6. Find the inverse of $S(t) = 95 - 12t$ and of $G(t) = 95(0.82)^t$.

7. Gaming drains 18% per hour. Find the equivalent continuous rate k in $G = 95e^{kt}$.

8. Explain why the streaming model eventually gives impossible values but the gaming model does not.

Part 3: Brightness & Rates

Day 3 · $L(t) = 400\cos\left(\frac{\pi}{12}(t - 13)\right) + 500$.

9. Find the average rate of change of L from 7 a.m. to 9 a.m. and from 12 p.m. to 2 p.m. Explain the difference.

10. Estimate when brightness changes fastest. Justify with the graph.

11. Write the lab report with a model for each phone feature and one question calculus could answer next.

Final Product: Battery Lab Report

Your final product must include:

- Charging analysis with difference quotients
- Composite temperature model
- Inverse and continuous-rate forms
- Brightness rates of change
- Lab report with next questions

Reflection

R1. How do difference quotients preview the derivative?

R2. Why choose exponential rather than linear models for charging?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Models for battery, charging, plans, and brightness are built and used correctly. Goes beyond the minimum with a creative, realistic choice.	Models for battery, charging, plans, and brightness are built and used correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Lab report includes models, graphs, calculations, and conclusions. Polished, organized, and visually engaging.	Lab report includes models, graphs, calculations, and conclusions. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.